

Digestive System

Midgut/Hindgut

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- derivatives of the midgut are :-
- duodenum distal to the opening of the bile duct, jejunum, ileum
- cecum, appendix, ascending colon & right two thirds of the transverse colon
- are supplied by the **superior mesenteric artery**,

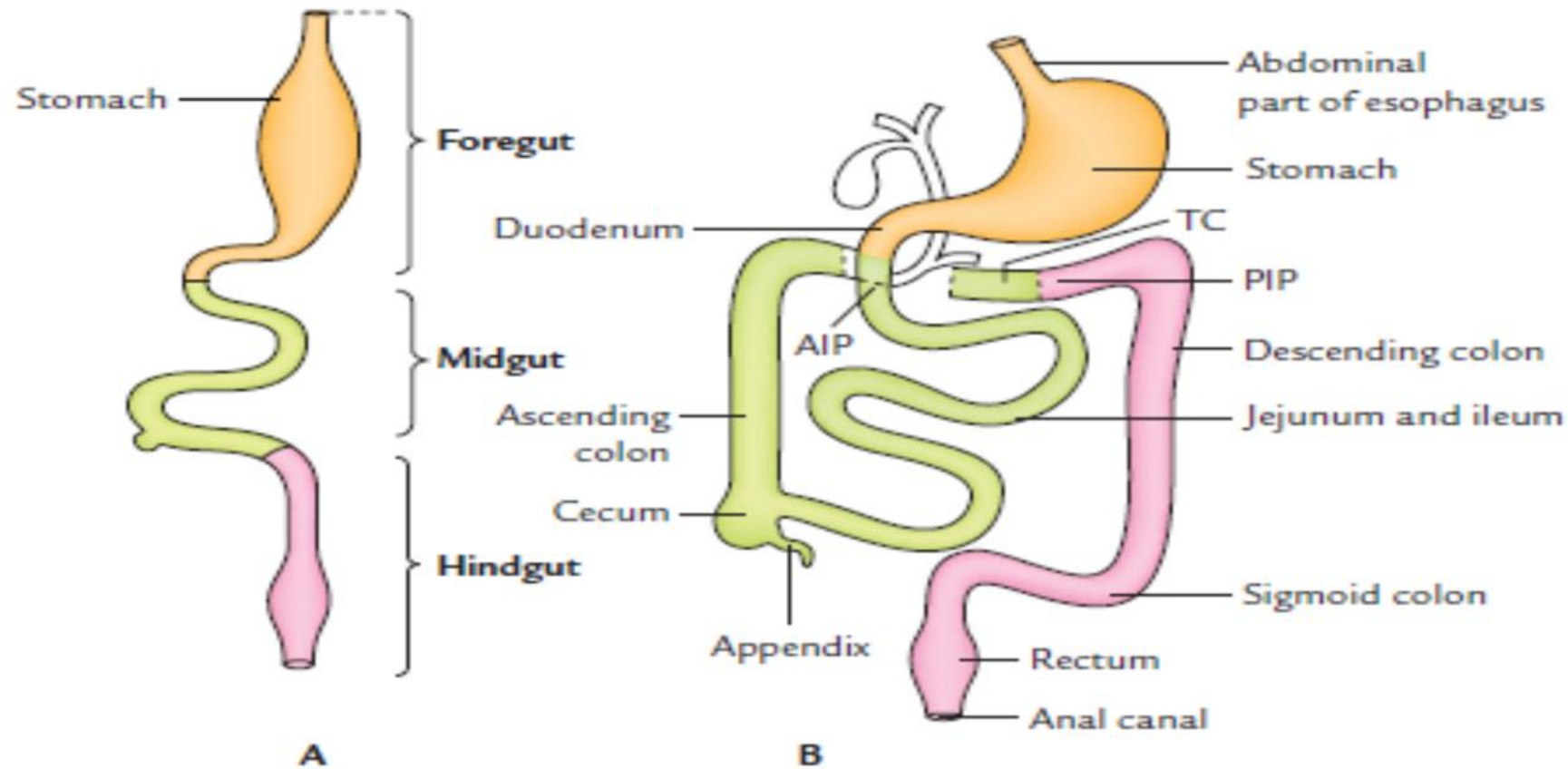


Fig. 13.5 Derivatives of various abdominal parts of the gut. A. Primitive gut. B. Adult gut. TC = transverse colon; AIP = anterior intestinal portal; PIP = posterior intestinal portal.

- midgut elongates forms a ventral, U-shaped loop of gut-**midgut loop of the intestine**
- projects into the remains of the extraembryonic coelom in the proximal part of the umbilical cord
- intraembryonic coelom communicates with extraembryonic coelom at the umbilicus
- midgut loop of the intestine **physiologic umbilical herniation** occurs at the beginning of the 6th week
- loop communicates with the umbilical vesicle through the narrow **omphaloenteric duct** (yolk stalk) until the 10th week

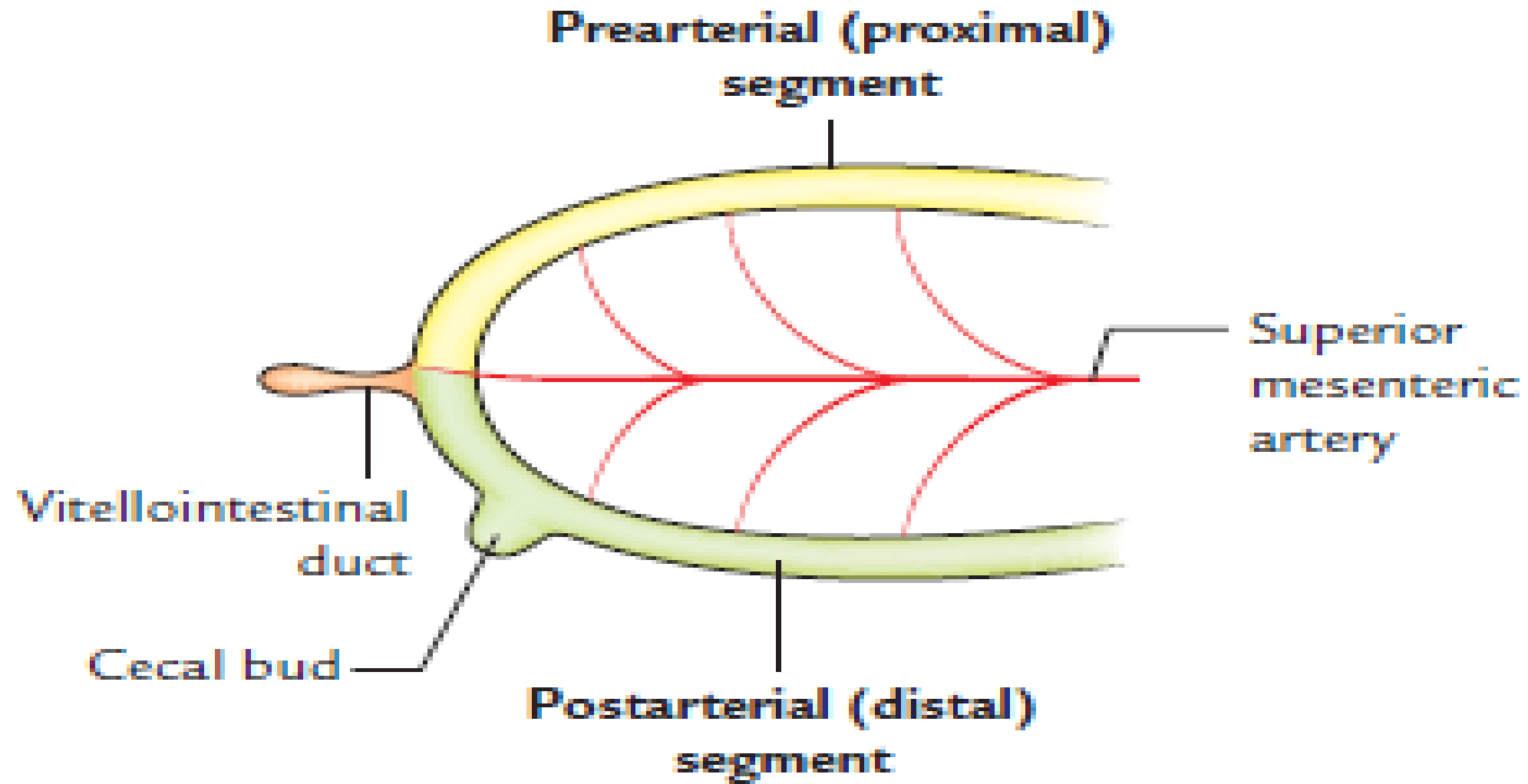
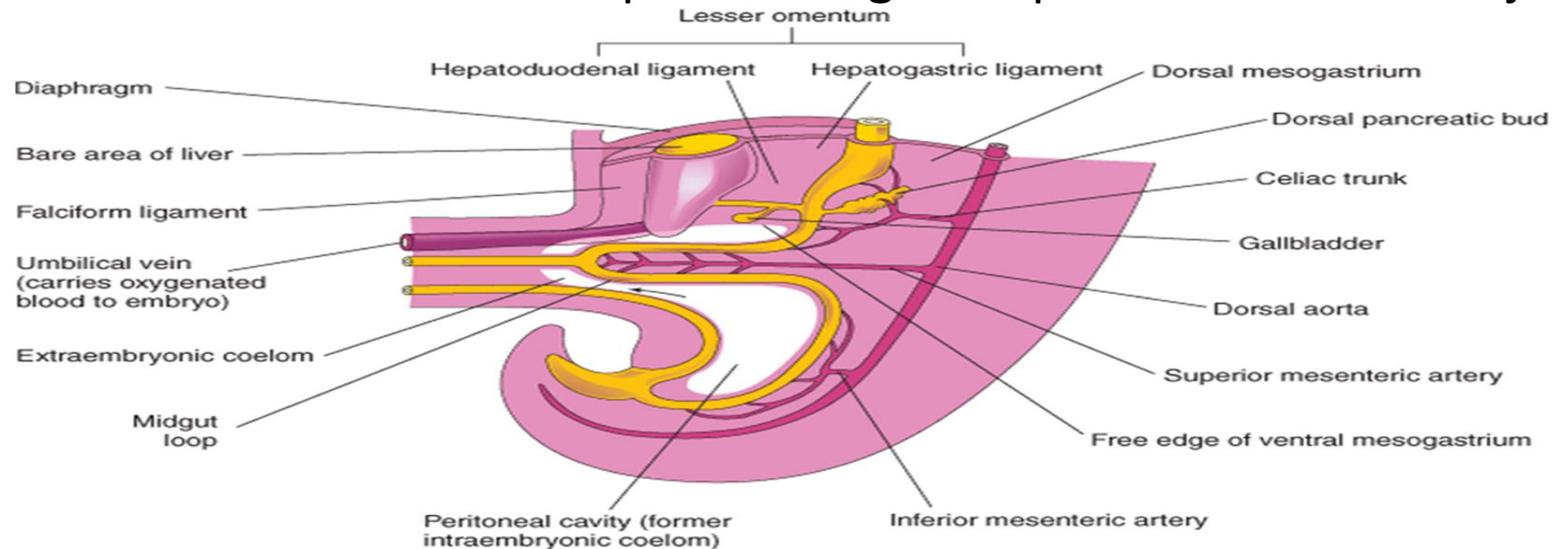
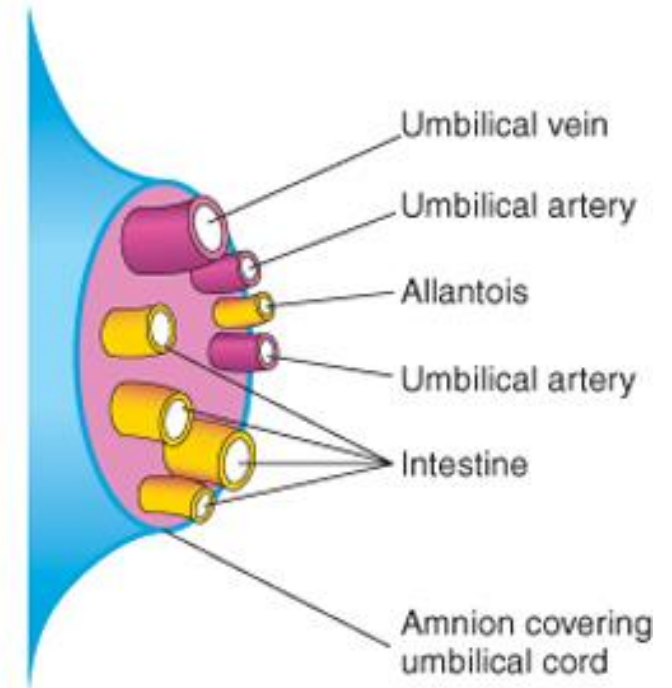
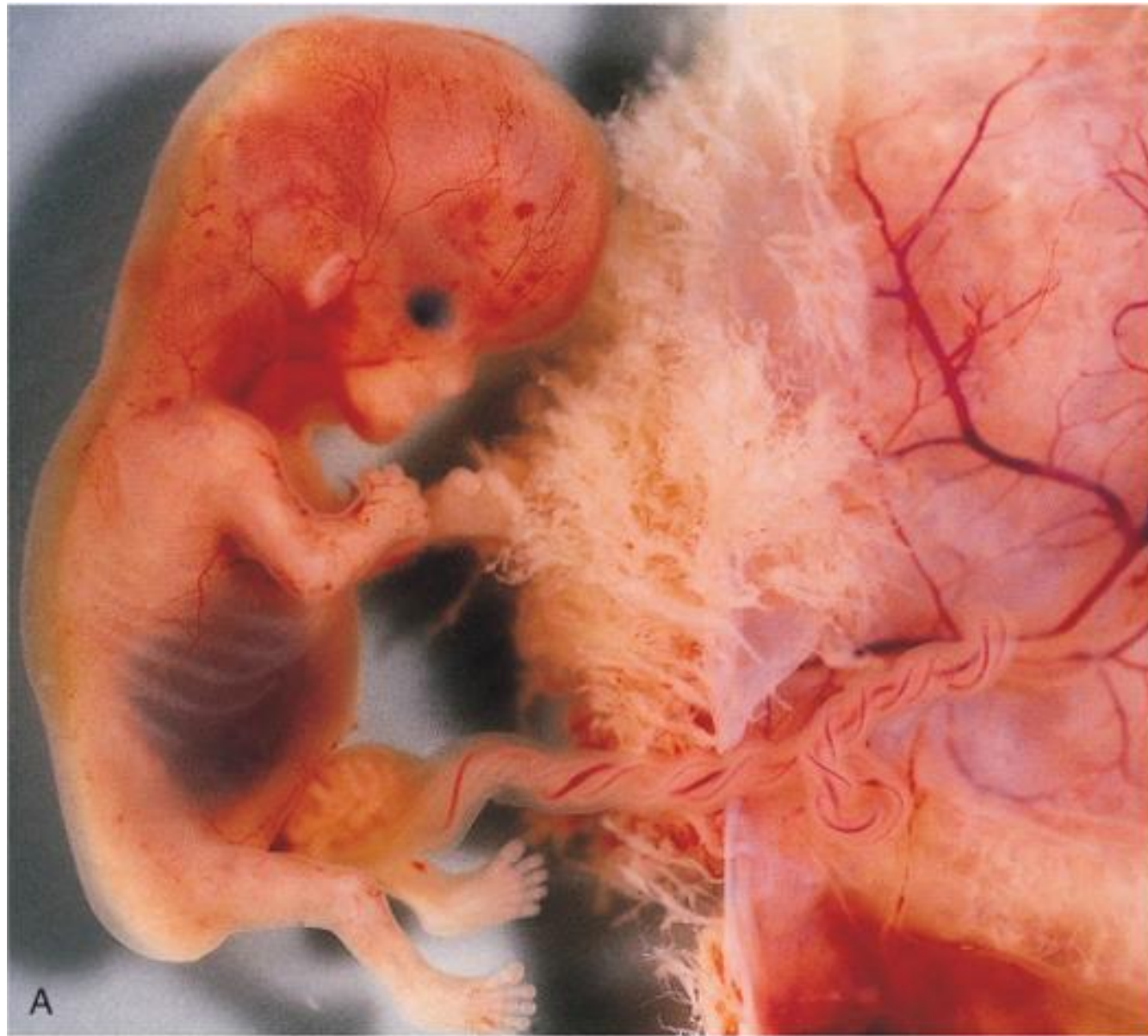


Fig. 13.15 Midgut loop.

- physiologic umbilical herniation occurs because there is not enough room in the abdominal cavity for the rapidly growing midgut
- shortage of space is caused mainly by relatively massive liver & kidneys that exist during this period of development
- midgut loop of intestine has a cranial (proximal) limb and a caudal (distal) limb
- suspended from the dorsal abdominal wall by an elongated mesentery
- omphaloenteric duct is attached to apex of midgut loop where the 2 limbs join





B

- cranial limb grows rapidly & forms small intestinal loops
- develops into the distal part of the duodenum, jejunum, & part of ileum
- caudal limb undergoes very little change except for development of the **cecal swelling** (diverticulum), primordium of cecum & appendix
- Becomes lower portion of ileum, cecum, appendix, ascending colon & proximal 2/3 of the transverse colon

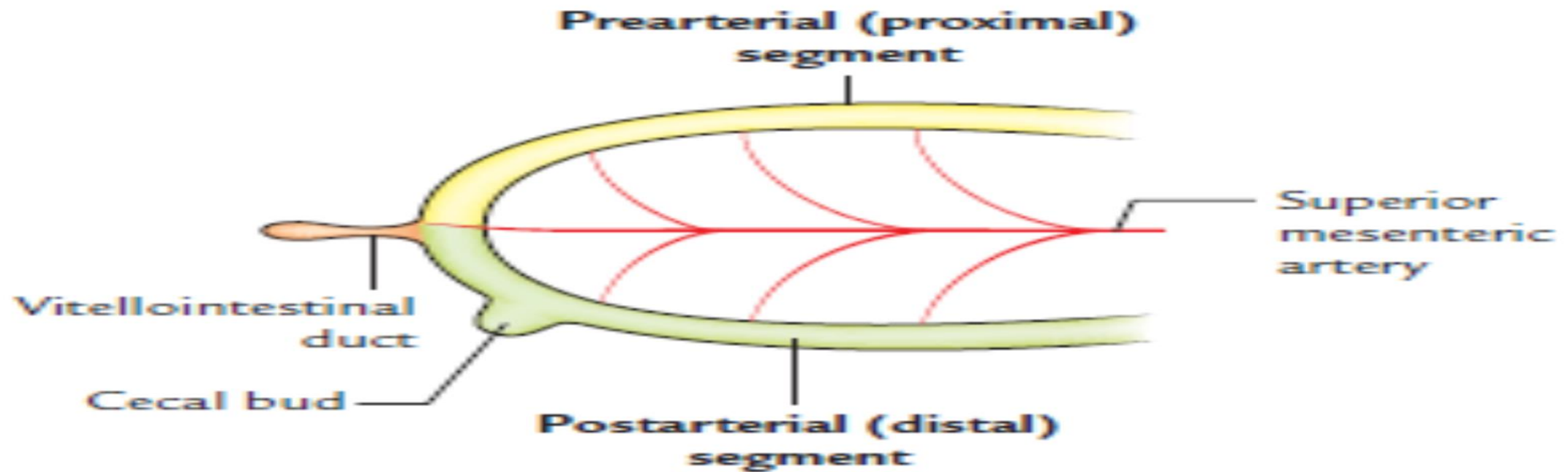
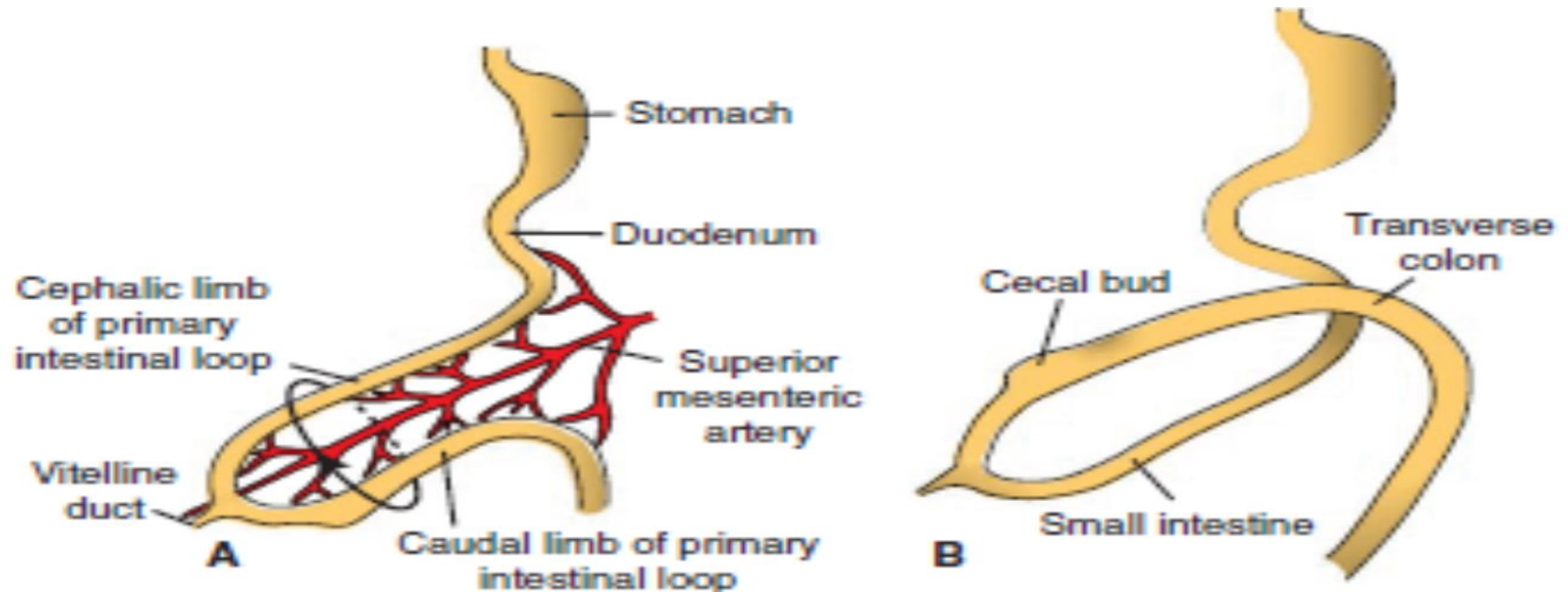
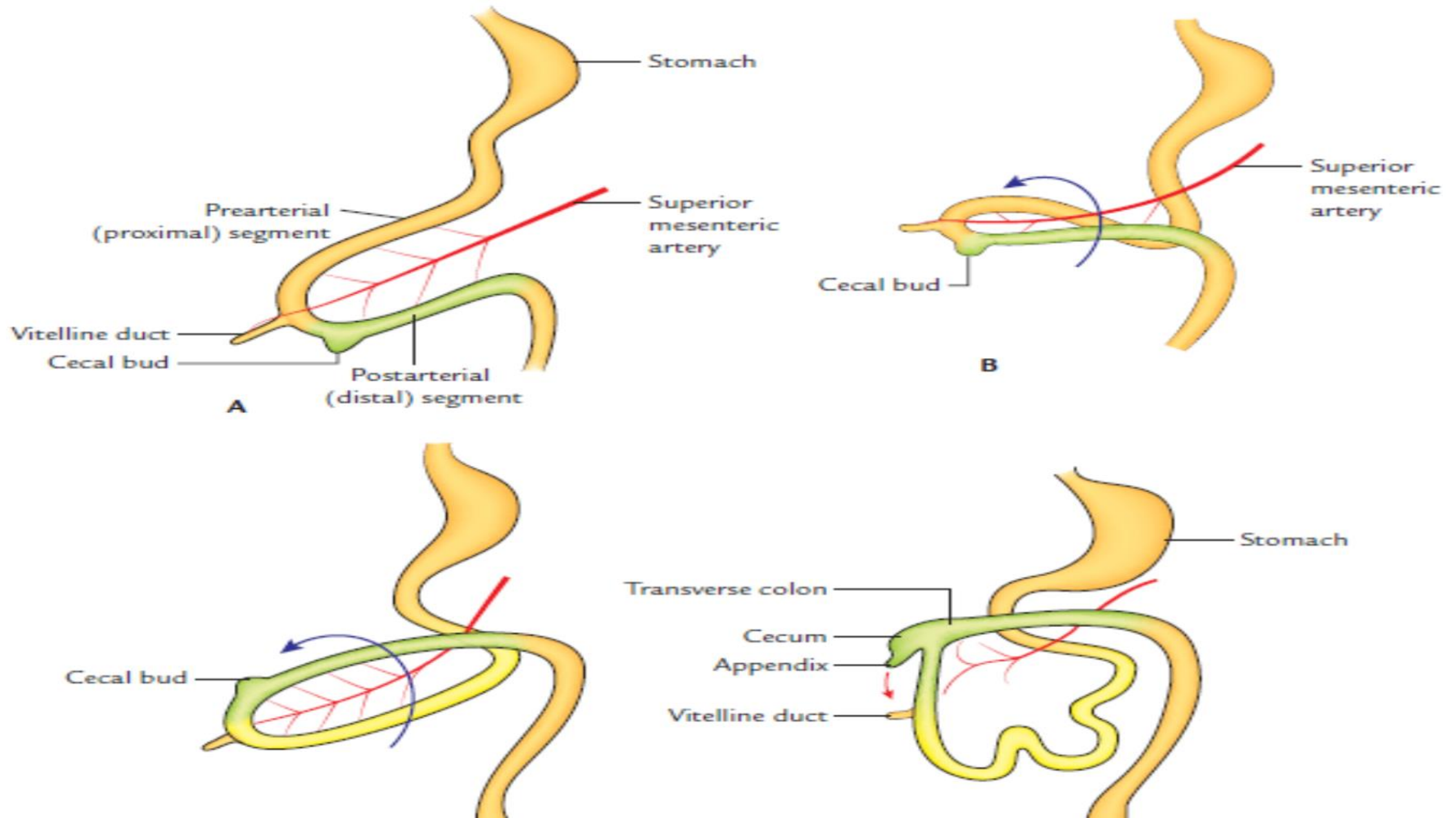


Fig. 13.15 Midgut loop.

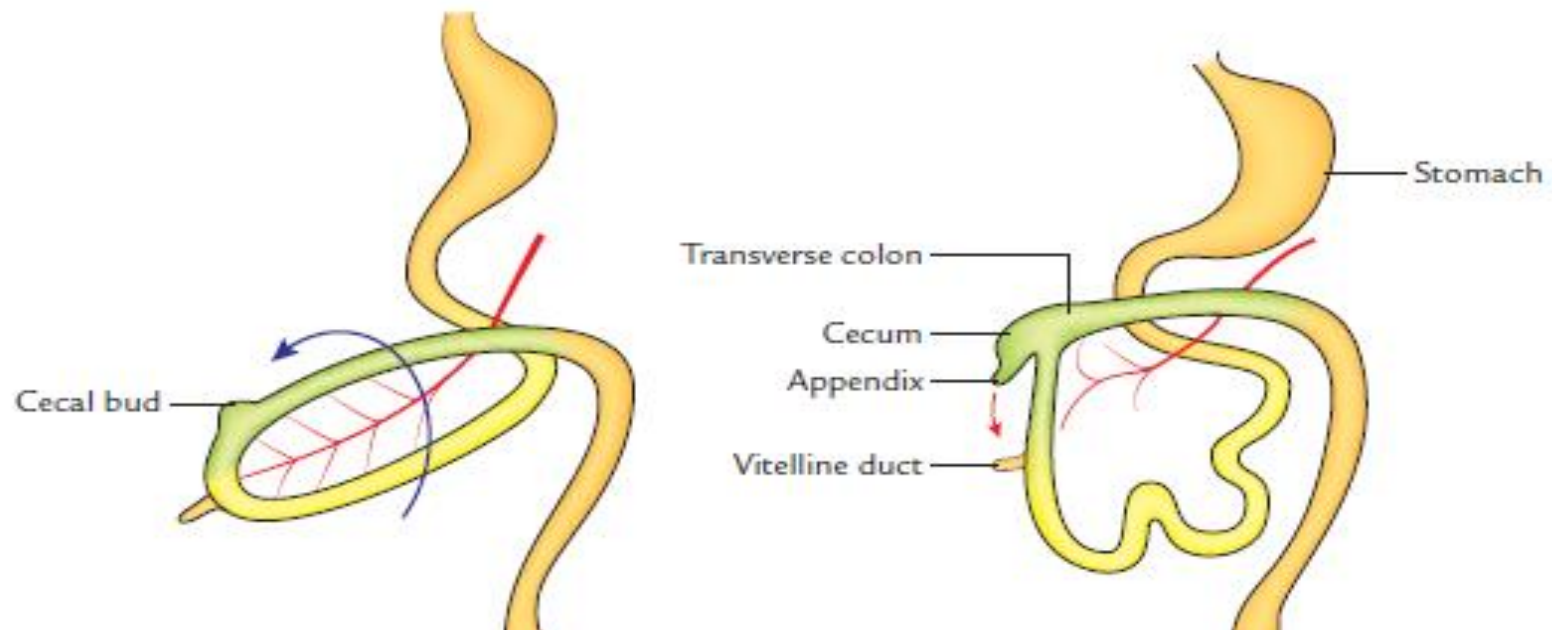
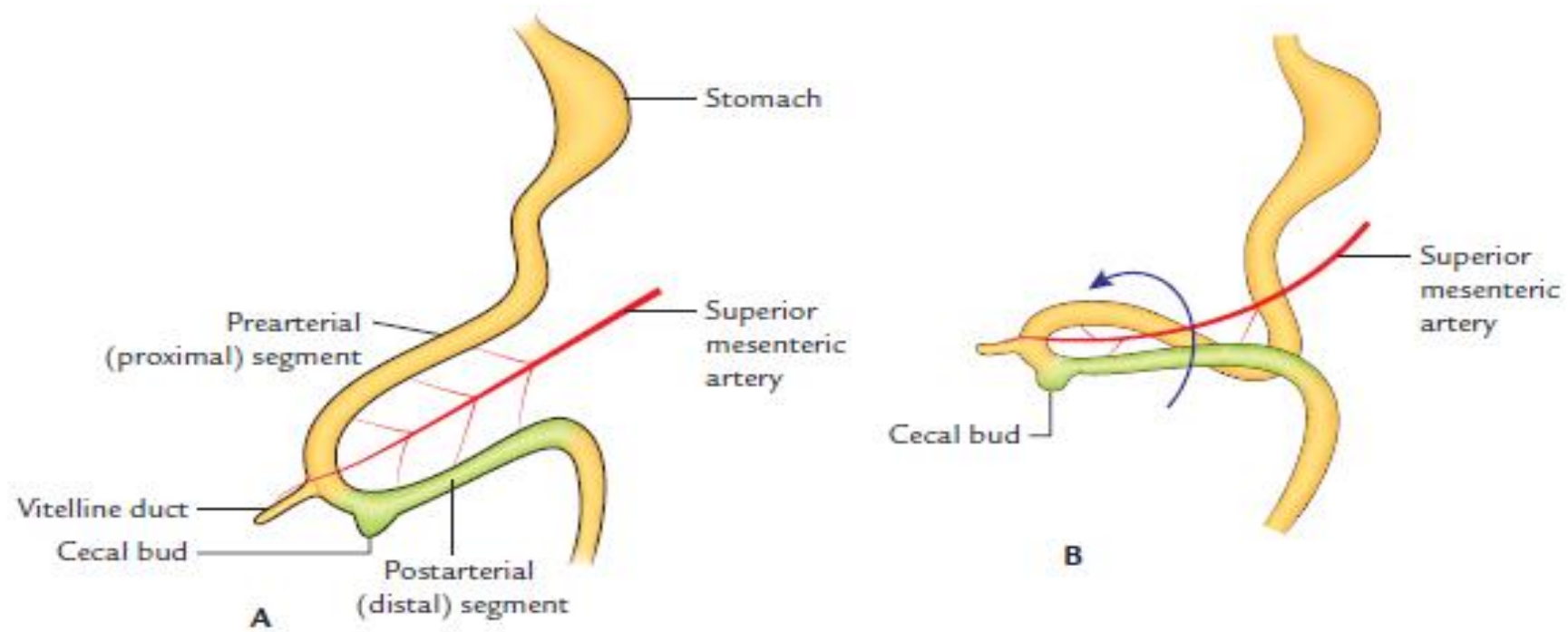
- **Rotation of the Midgut Loop**
- While in the umbilical cord, midgut loop rotates 90 degrees counterclockwise (looking from the ventral side) around the axis of the **superior mesenteric artery**
- brings cranial limb (small intestine) of the midgut loop to the right & caudal limb (large intestine) to the left
- During rotation, cranial limb elongates & forms intestinal loops (e.g. primordia of jejunum & ileum).



- Rotation plays a very important part in establishing the definitive relationships of the various parts of the intestine



- **Return of the Midgut to the Abdomen**
- 10th week, intestines return to the abdomen (reduction of the midgut hernia)
- enlargement of the abdominal cavity & relative decrease in the size of liver & kidneys are important factors to the return
- small intestine (formed from the cranial limb) returns first turn 90 degrees
- pass posterior to superior mesenteric artery & occupies the central part of the abdomen
- large intestine also return undergoes a further 90-degree counterclockwise rotation
- Later it comes to occupy the right side of the abdomen
- ascending colon becomes recognizable as the posterior abdominal wall elongates



- **Fixation of the Intestines**

- Rotation of the stomach and duodenum causes the duodenum and pancreas to fall to the right
- enlarged colon presses the duodenum and pancreas against the posterior abdominal wall
- results most of the duodenal mesentery to be absorbed
- the duodenum, except for approximately the first 2.5 cm (derived from foregut), has no mesentery and lies retroperitoneally
- the head of the pancreas becomes retroperitoneal (posterior to peritoneum)

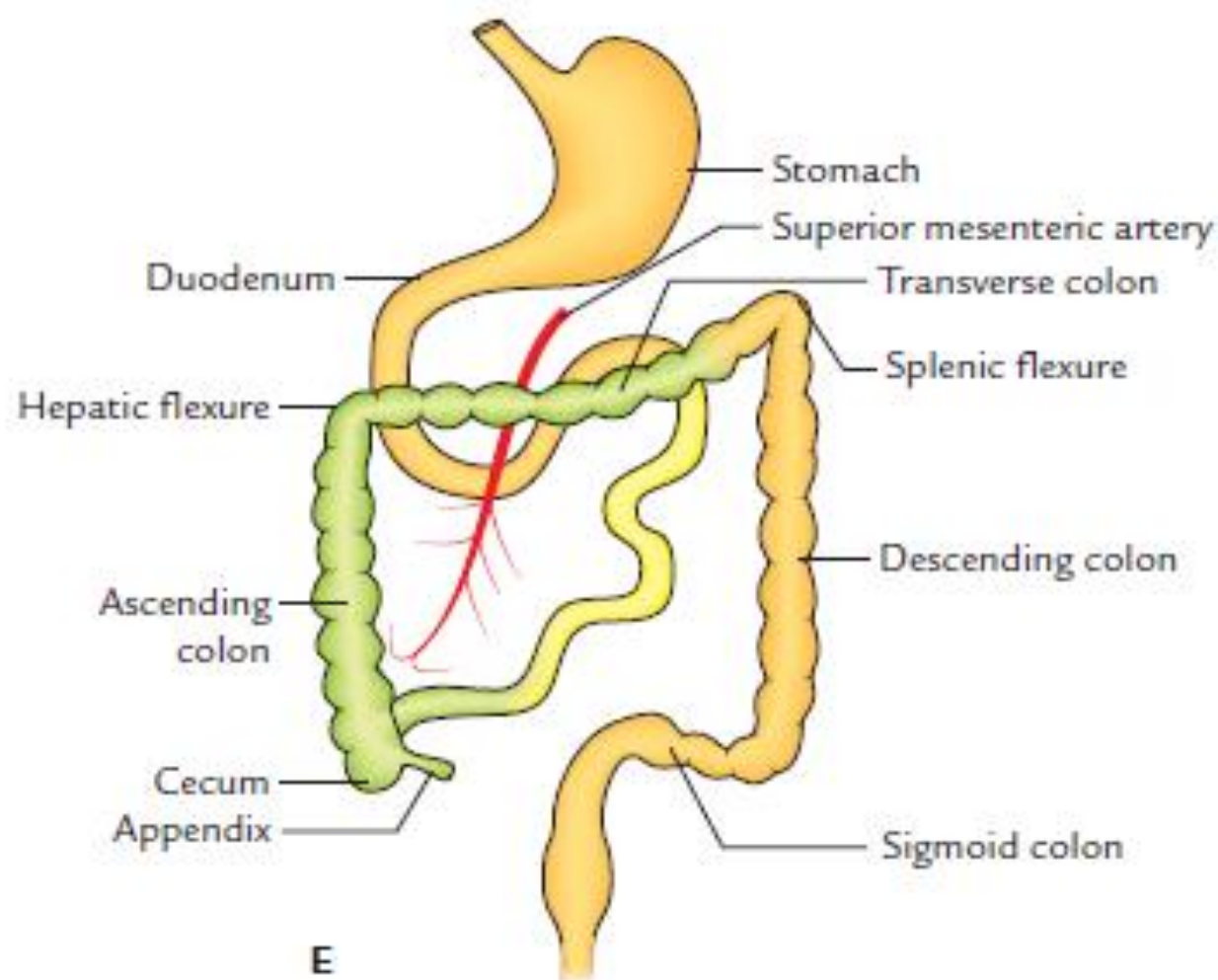


Fig. 13.16 Rotation of midgut loop as seen in left side view. A. Primitive loop before rotation. B. Anticlockwise 90° rotation of midgut loop while it is in the extraembryonic celom in the umbilical cord. C. Anticlockwise 180° rotation of midgut loop as it is withdrawn into the abdominal cavity. D. Descent of cecum takes place later. E. Intestinal loops in final position.

- intestines enlarge, lengthen & assume their final positions, their mesenteries are pressed against the posterior abdominal wall
- mesentery of the ascending colon fuses with the parietal peritoneum on this wall disappears
- consequently, ascending colon also becomes retroperitoneal
- jejunum & ileum retain their mesenteries
- fan-shaped mesentery of the small intestines acquires a new line of attachment
- passes from the duodenojejunal junction inferolaterally to the ileocecal junction

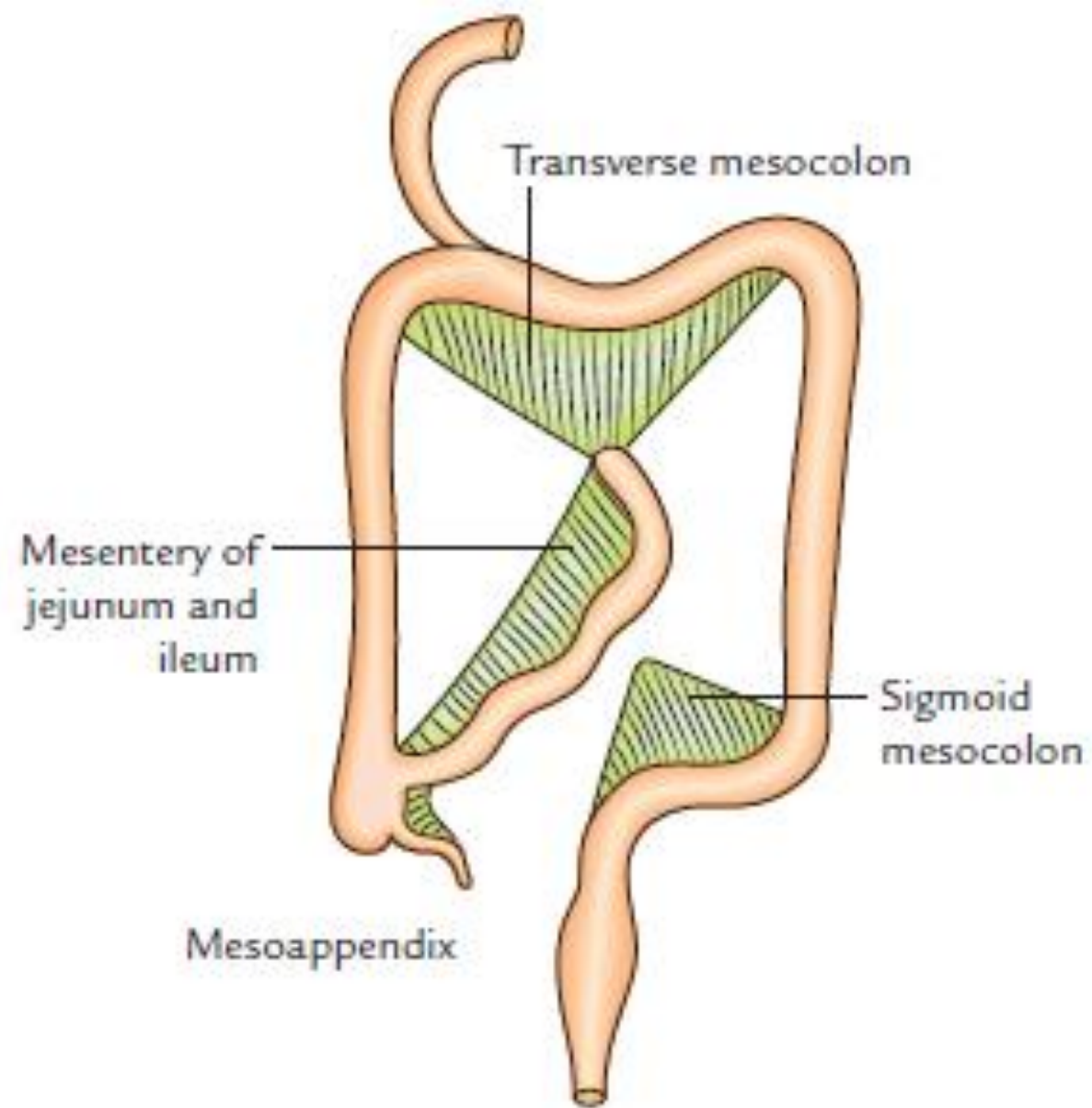


Fig. 13.30 Fate of dorsal mesentery of midgut and hindgut.

- **The Cecum and Appendix**

- primordium of the cecum & appendix- **cecal swelling** (diverticulum)-appears in the 6th week as an elevation on the antimesenteric border of the caudal limb of the midgut loop
- apex of the cecal swelling does not grow as rapidly as the rest of it
- appendix is initially a small diverticulum of the cecum
- appendix increases rapidly in length so that at birth it is a relatively long tube arising from the distal end of the cecum
- After birth, the wall of the cecum grows unequally, with the result that the appendix comes to enter its medial side

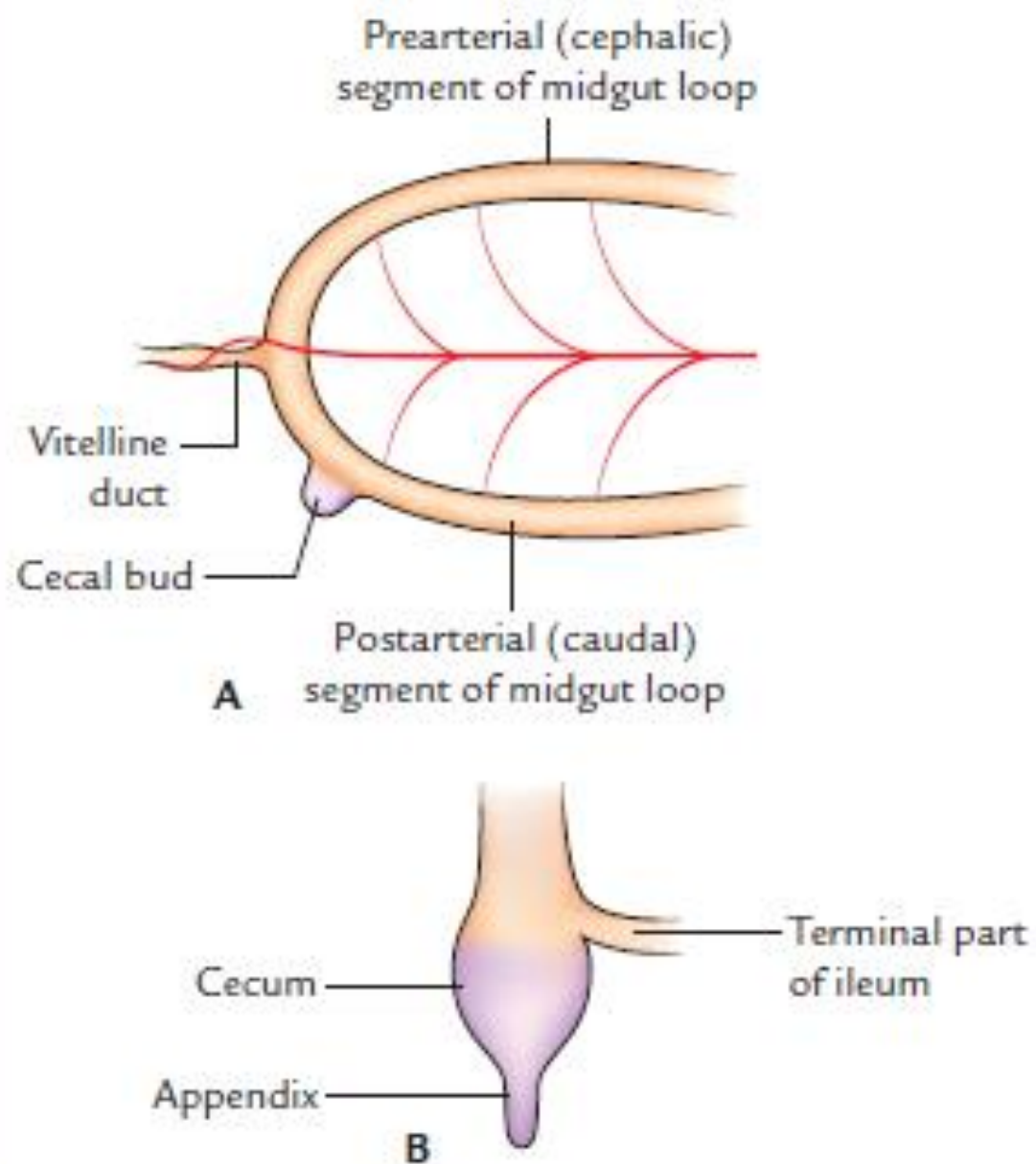


Fig. 13.18 Development of cecum.

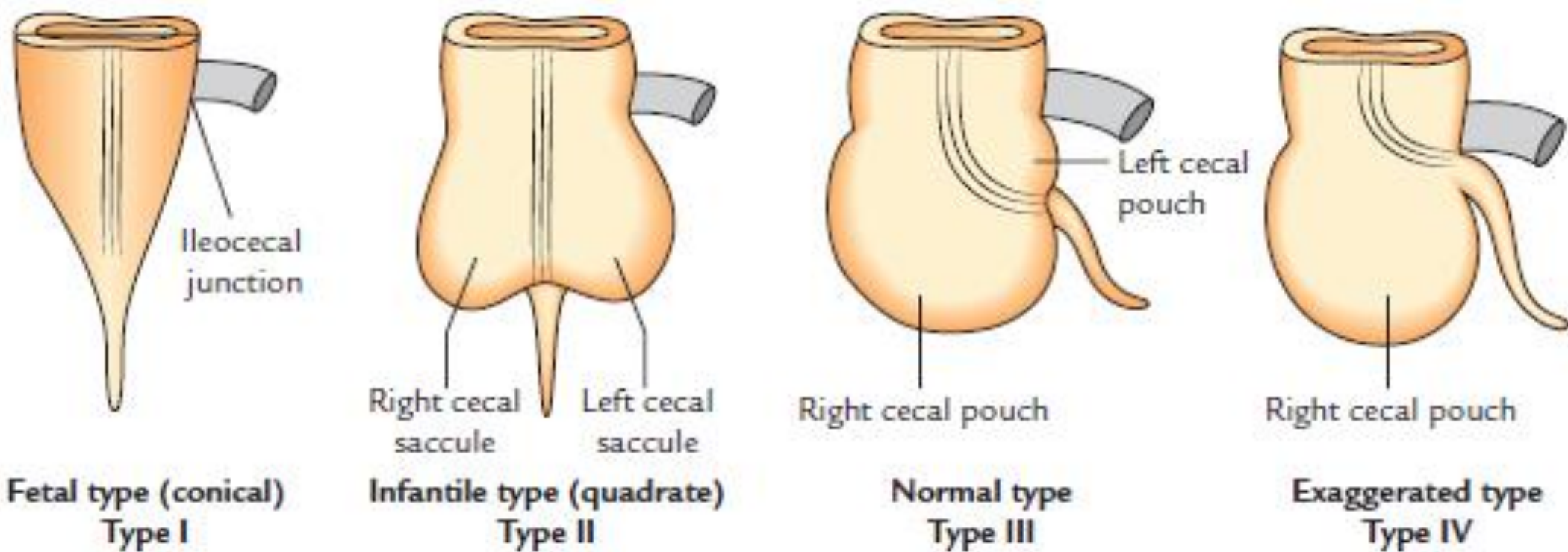
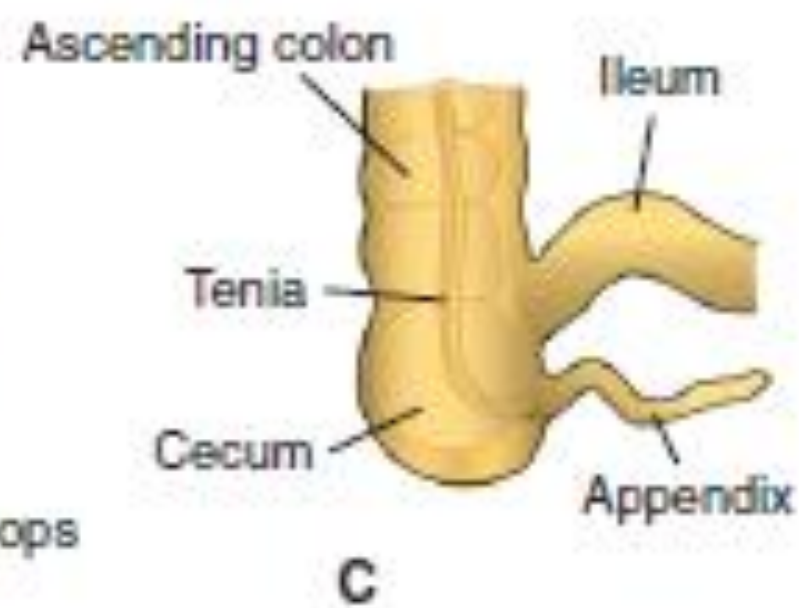
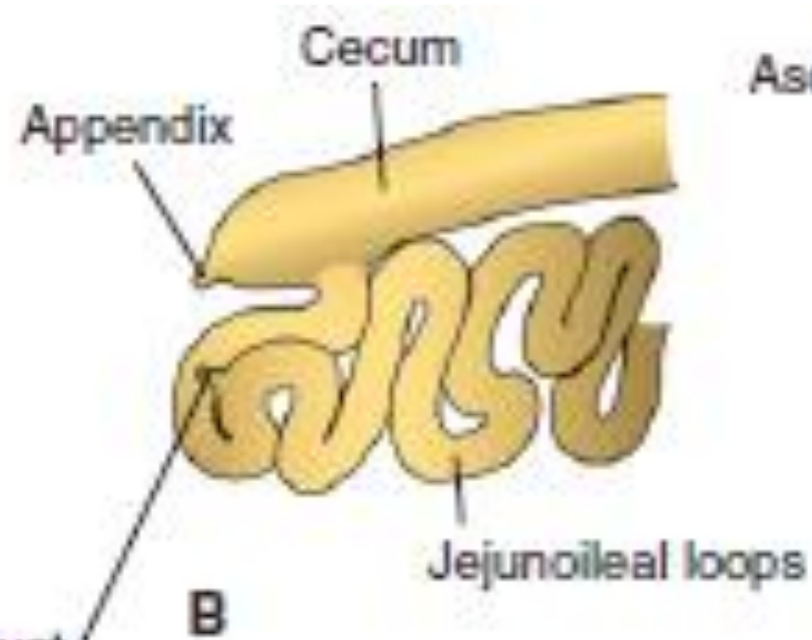


Fig. 13.19 Types of cecum.



• HINDGUT

- derivatives of the hindgut are:-
- left one third to one half of the transverse colon,
- descending colon & sigmoid colon,
- Rectum & the superior part of the anal canal
- derivatives are supplied by the **inferior mesenteric artery**
- junction between segment of transverse colon from midgut & hindgut is indicated by the change in blood supply from superior to inferior mesenteric artery

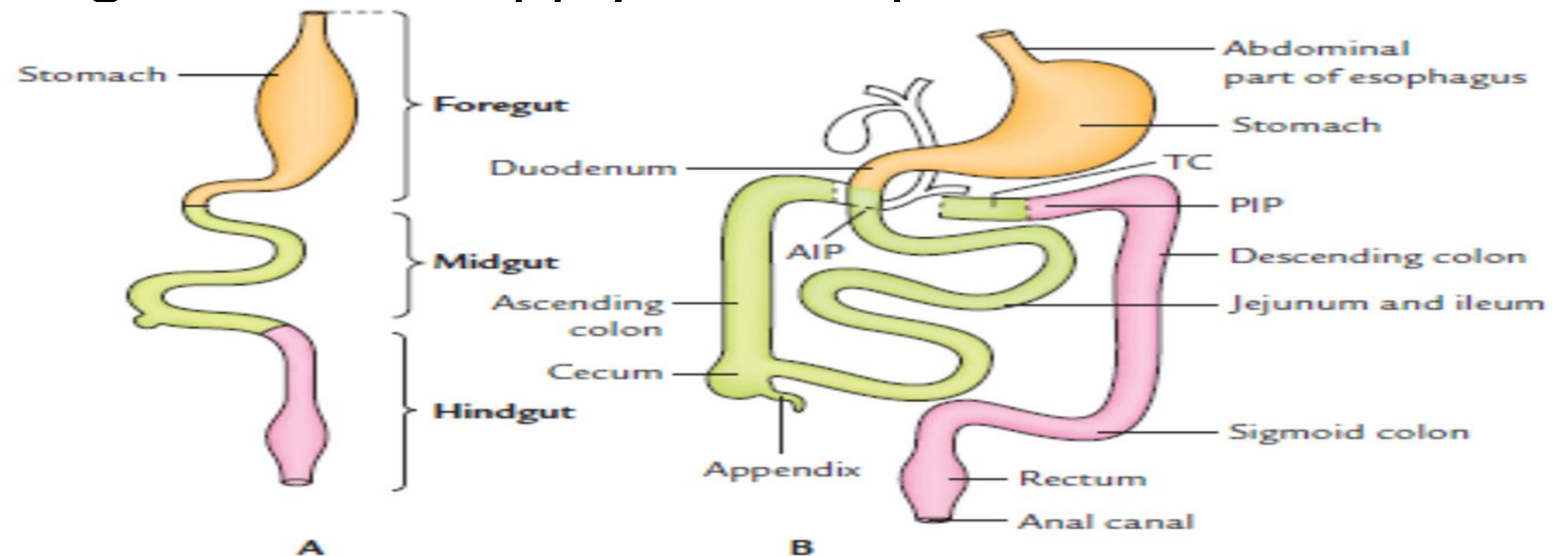
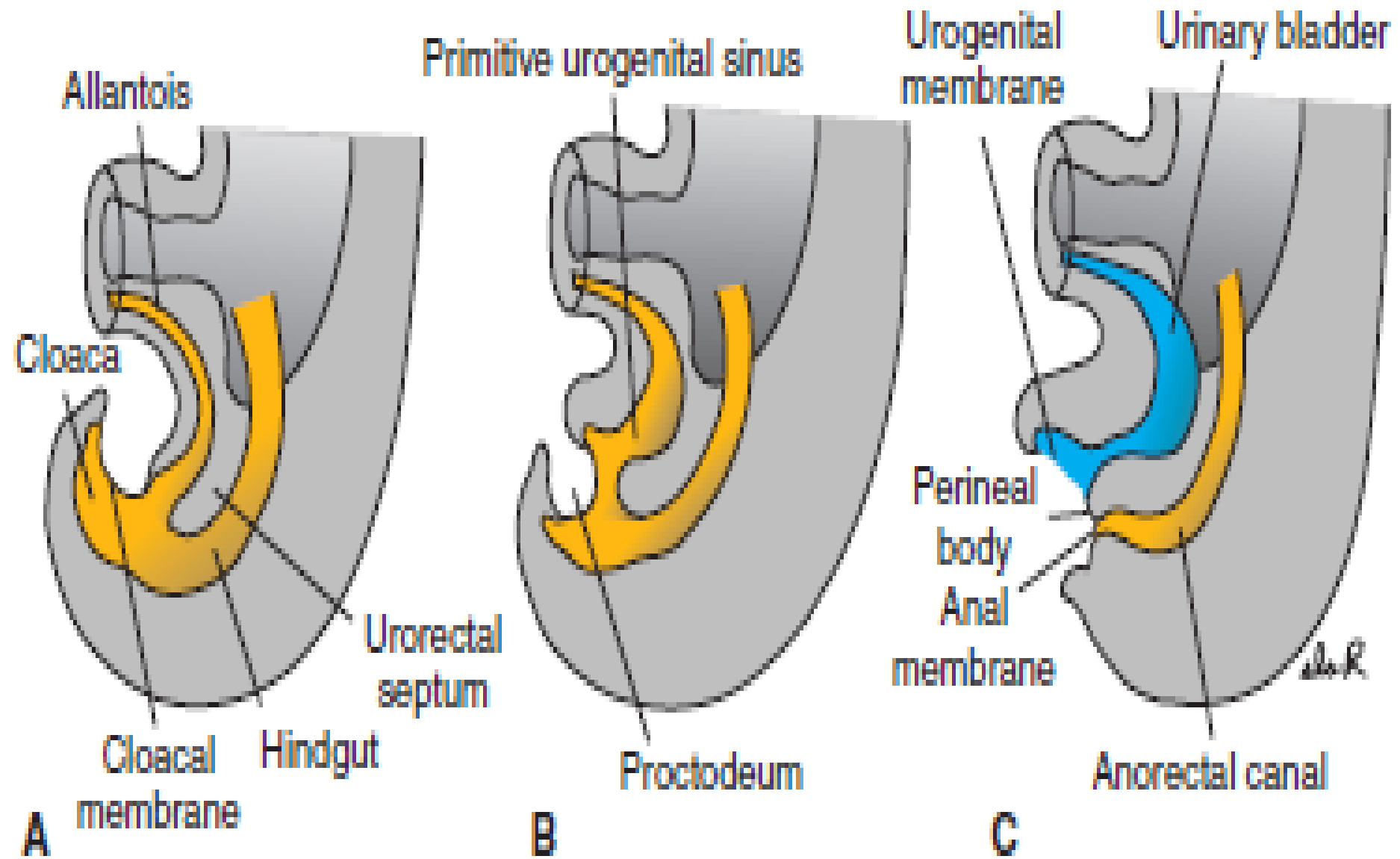


Fig. 13.5 Derivatives of various abdominal parts of the gut. A. Primitive gut. B. Adult gut. TC = transverse colon; AIP = anterior intestinal portal; PIP = posterior intestinal portal.

- descending colon becomes retroperitoneal as its mesentery fuses with the peritoneum on the left posterior abdominal wall and then disappears
- mesentery of the sigmoid colon is retained
- **Cloaca**
- expanded terminal part of the hindgut is an endoderm-lined chamber
- is in contact with the surface ectoderm at the **cloacal membrane**
- membrane is composed of endoderm of the cloaca and ectoderm of the **proctodeum** or anal pit



- cloaca receives **allantois** ventrally which is a fingerlike diverticulum
- **Partitioning of the Cloaca**
- cloaca is divided into dorsal & ventral parts by a wedge of mesenchyme-the **urorectal septum**
- develops in the angle between the allantois & hindgut
- septum grows toward the cloacal membrane
- develops forklike extensions that produce infoldings of the lateral walls of the cloaca

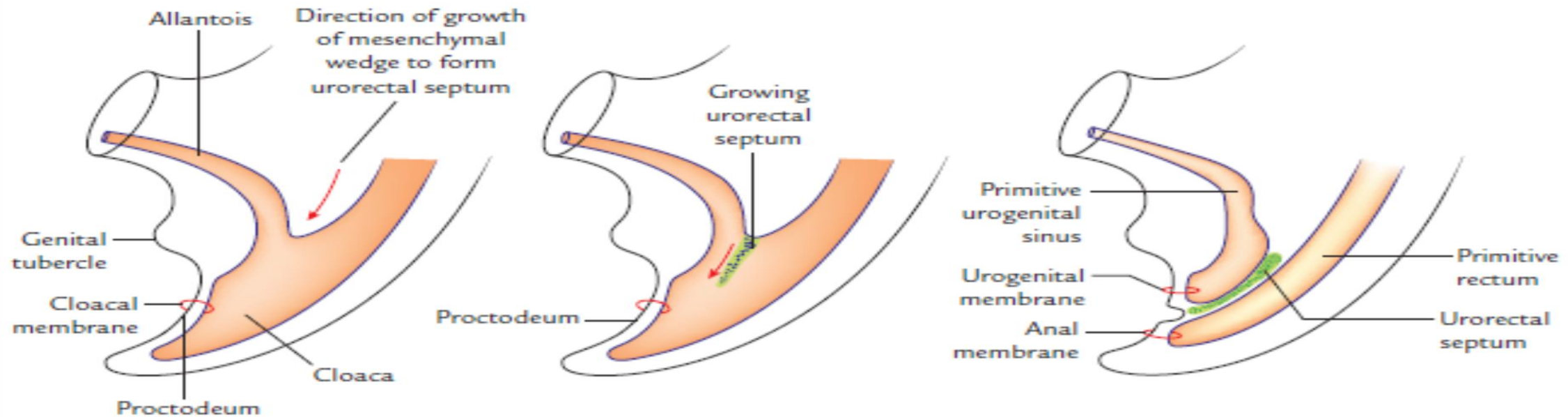


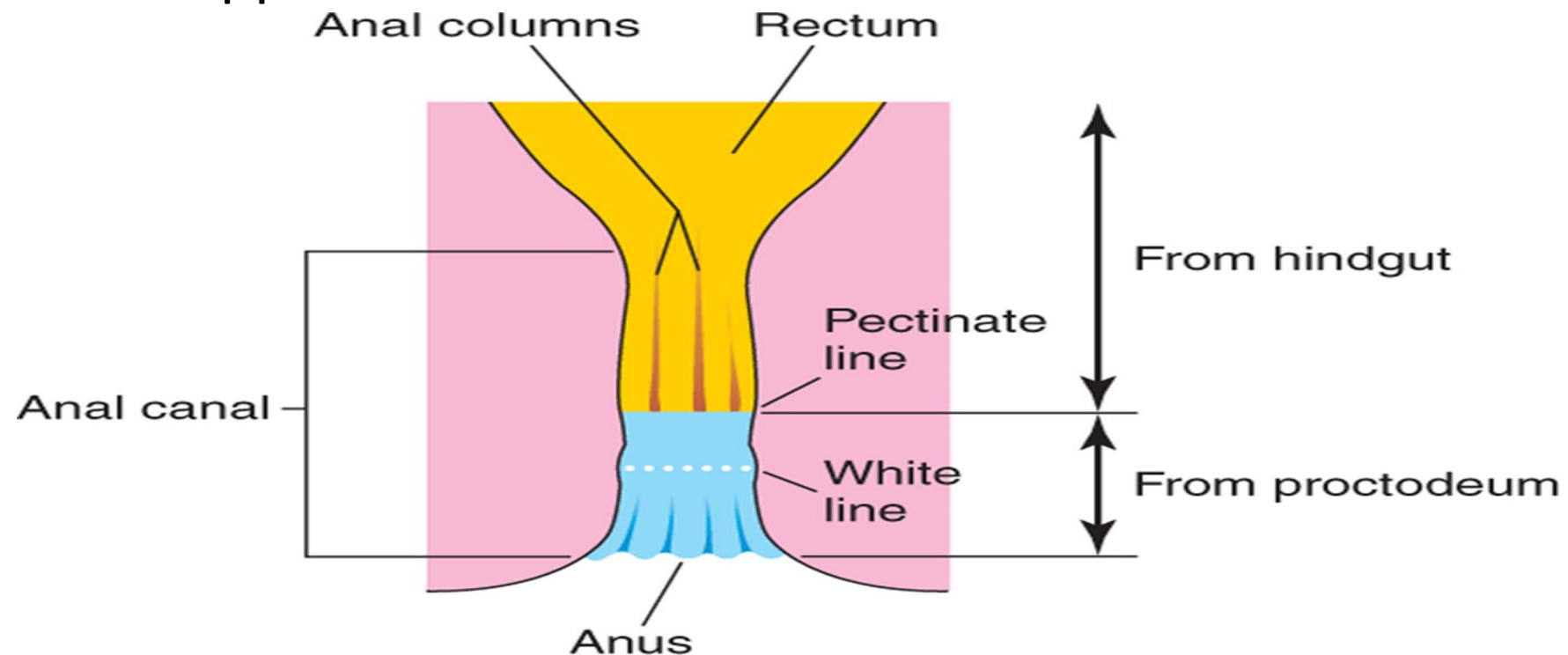
Fig. 13.25 Successive stages of formation of urorectal septum, which divides the cloaca into anterior part (the primitive urogenital sinus) and posterior part (the primitive rectum).

- folds grow toward each other & fuse, forming a partition that divides the cloaca into 2 parts
 - ❖ *rectum* & cranial part of the *anal canal* dorsally
 - ❖ *urogenital sinus* ventrally
- 7th week urorectal septum has fuses with the cloacal membrane
- dividing it into a dorsal **anal membrane** and a larger ventral **urogenital membrane**
- area of fusion of the urorectal septum with the cloacal membrane is represented in the adult by the **perineal body**, the tendinous center of the perineum

- fibromuscular node of the perineum is where several muscles converge and attach
- urorectal septum also divides the cloacal sphincter into anterior and posterior parts
- posterior part becomes the external anal sphincter,
- anterior part develops into the superficial transverse perineal, bulbospongiosus, and ischiocavernosus muscles
- This developmental fact explains why one nerve, the *pudendal nerve*, supplies all these muscles

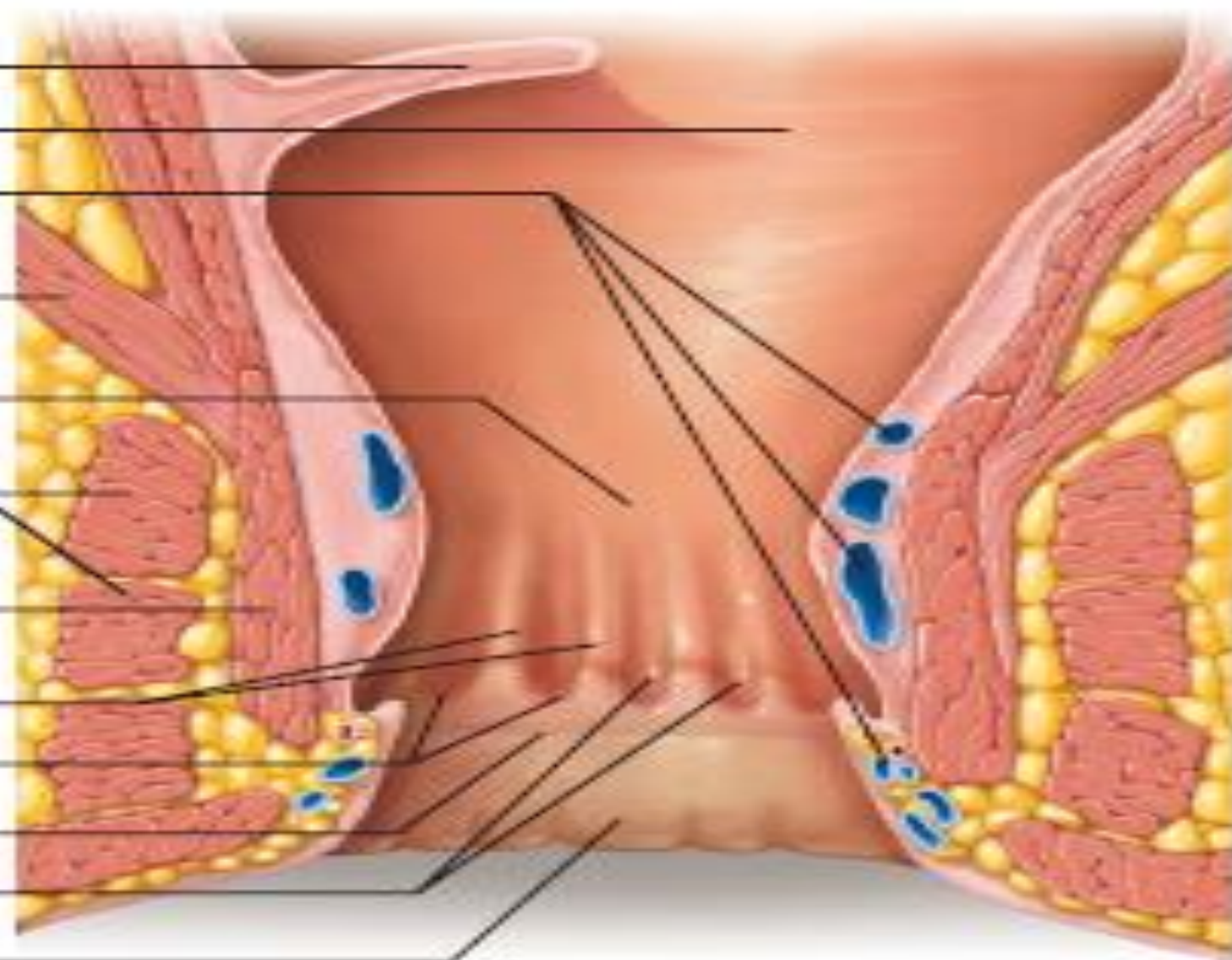
- Mesenchymal proliferations produce elevations of the surface ectoderm around the **anal membrane**
- membrane is soon located at the bottom of an ectodermal depression-the **proctodeum** or anal pit
- anal membrane usually ruptures at the end of the eighth week
- brings the distal part of the digestive tract (anal canal) into communication with the amniotic cavity

- **The Anal Canal**
- superior 2/3 of the adult anal canal are derived from the **hindgut**
- inferior 1/3 develops from the **proctodeum**
- junction of the epithelium derived from the ectoderm of Proctodeum & endoderm of hindgut is roughly indicated by the irregular **pectinate line** located at the inferior limit of anal valves
- line indicates the approximate former site of anal membrane



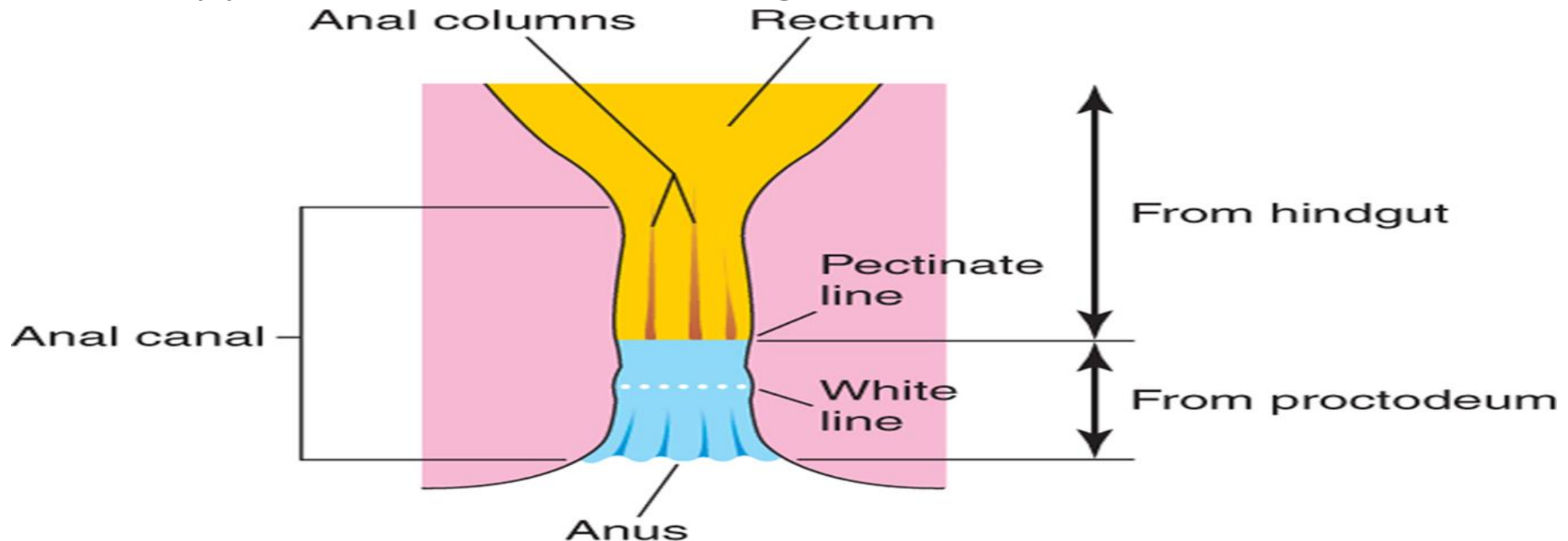
(a)

- Rectal valve
- Rectum
- Hemorrhoidal veins
- Levator ani muscle
- Anal canal
- External anal sphincter
- Internal anal sphincter
- Anal columns
- Anal valves
- Pectinate line
- Anal sinuses
- Anus



(b)

- 2 cm superior to anus is an **anocutaneous line** ("white line")
- where the composition of the anal epithelium changes from columnar to stratified squamous cells
- At the anus, epithelium is keratinized & continuous with the skin around the anus
- other layers of the wall of anal canal are derived from splanchnic mesenchyme
- pyloric sphincter & the ileocecal valve (sphincter), the formation of the anal sphincter appears to be under Hox D genetic control



- superior 2/3 of anal canal are supplied by *superior rectal artery*, branch of inferior mesenteric artery due to its hindgut origin
- venous drainage of this superior part is mainly via the *superior rectal vein*
- lymphatic drainage of the superior part is eventually to the *inferior mesenteric lymph nodes*
- Its nerves are from the autonomic nervous system
- its origin from the proctodeum, the inferior 1/3 of anal canal is supplied mainly by *inferior rectal arteries*, branches of the internal pudendal artery

- venous drainage is through the *inferior rectal vein*, a tributary of the internal pudendal vein that drains into the internal iliac vein
- *lymphatic drainage* of the inferior part of the anal canal is to the *superficial inguinal lymph nodes*.
- nerve supply is from the *inferior rectal nerve*; hence, it is sensitive to pain, temperature, touch, and pressure

- differences in blood supply, nerve supply, and venous and lymphatic drainage of the anal canal are important clinically
- e.g. when considering the metastasis (spread) of cancer cells
- characteristics of carcinomas in the 2 parts also differ
- Tumors in the superior part are painless & arise from columnar epithelium,
- whereas those in the inferior part are painful & arise from stratified squamous epithelium.

- ***Oesophageal atresia:***
- occurs due to failure of recanalization of the developing oesophagus
- associated with **tracheoesophageal fistula**
- produced by extreme posterior deviation of **tracheoesophageal septum**
- foetus is unable to swallow amniotic fluid
- there is an abnormal increase in the amount of amniotic fluid producing a clinical condition called **polyhydramnios**
- surgical correction (treatment) gives 85% survival rate

- ***Oesophageal stenosis:***
- lumen of the oesophagus is narrow usually in *lower third part*
- caused by incomplete oesophageal recanalization and vascular abnormalities
- ***Achalasia cardia:*** occurs due to failure of relaxation of the musculature in the lower part of the oesophagus following loss of ganglionic cells in Auerbach's plexus
- ***Exomphalos or omphalocele*** results from failure of coils of the small intestine to return into abdominal cavity from their physiological herniation into extraembryonic celom during 6th to 10th week of IUL
- mass contains coils of the small intestine and is covered by a transparent amniotic membrane

- ***Congenital umbilical hernia:***

- Herniation of abdominal viscera through the weak umbilical opening (poorly closed umbilicus)
- presents as a protrusion in the linea alba
- contents are covered with peritoneum, subcutaneous tissue, and skin

- ***Gastroschisis:***

- there is a linear defect in anterior abdominal wall through which abdominal contents herniate out
- occurs lateral to the umbilicus, usually on to the right
- defect is produced when lateral folds of embryo fail to fuse with each other around connecting stalk

- *Meckel's diverticulum* small part of vitellointestinal duct close to midgut (ileum) persists and forms the Meckel's diverticulum
- It may be connected to the umbilicus by a fibrous
- ***Congenital megacolon (Hirschsprung's disease)***
- a segment of the colon is dilated
- Is the segment distal to dilatation that is abnormal
- this abnormal segment, autonomic parasympathetic ganglia are absent in the myenteric plexus
- there is no peristalsis in this segment.
- Since contents of colon cannot pass through this segment, the segment proximal to it grossly dilates
- commonly seen in the sigmoid colon or rectum

- **Volvulus**

- parts of intestine that normally become retroperitoneal may retain mesentery
- As a result, they become highly mobile due to hypermotility—a portion of intestine twist along with its blood vessels on the axis of mesentery
- Consequently the blood supply is compromise

- Congenital Omphalocele
- Meckel's diverticulum
- Umbilical Hernia
- Gastroschisis
- Anomalies of the Midgut(nonrotation and malrotation)
- Reversed Rotation
- Subhepatic Cecum and Appendix
- Internal Hernia
- Stenosis and Atresia of the Intestine
- Ileal Diverticulum and Other Omphaloenteric Remnants

- Anomalies of the Hindgut
- Congenital Megacolon or Hirschsprung Disease
- Imperforate Anus and Anorectal Anomalies
- Anal Agenesis, with or without a Fistula
- Anal Stenosis
- Situs inversus
- Congenital anomalies due to errors of fixation of the gut
- Rectal fistulae